PRAKTIKUM BASIS DATA

MODUL 8

SCRIPTING SQL (CREATE INDEX, VIEW, ANALYZE TABLE)

Disusun oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

FAKULTAS INFORMATIKA

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JURNAL MODUL 8

1. buat tabel database untuk suatu perusahaan

Tabel Departments

```
CREATE TABLE departments (
    department_id NUMBER PRIMARY KEY,
    department_name VARCHAR2(100),
    manager_id NUMBER,
    location_id NUMBER
);
```

Tabel Employees

```
CREATE TABLE employees (
    employee_id NUMBER PRIMARY KEY,
    last_name VARCHAR2(100) NOT NULL,
    email VARCHAR2(100),
    salary NUMBER,
    commission_pct NUMBER,
    hire_date DATE NOT NULL,
    department_id NUMBER,
    CONSTRAINT fk_dept FOREIGN KEY (department_id)
        REFERENCES departments(department_id)
);
```

Tabel Suppliers

```
CREATE TABLE suppliers (
    supplier_id NUMBER PRIMARY KEY,
    supplier_name VARCHAR2(100),
    contact VARCHAR2(100)
);
```

Tabel Products

```
CREATE TABLE products (
    product_id NUMBER PRIMARY KEY,
    product_name VARCHAR2(100),
    stock NUMBER,
    supplier_id NUMBER,
    CONSTRAINT fk_supplier FOREIGN KEY (supplier_id)
        REFERENCES suppliers(supplier_id)
);
```

2. Analisis table menggunakan compute statistic

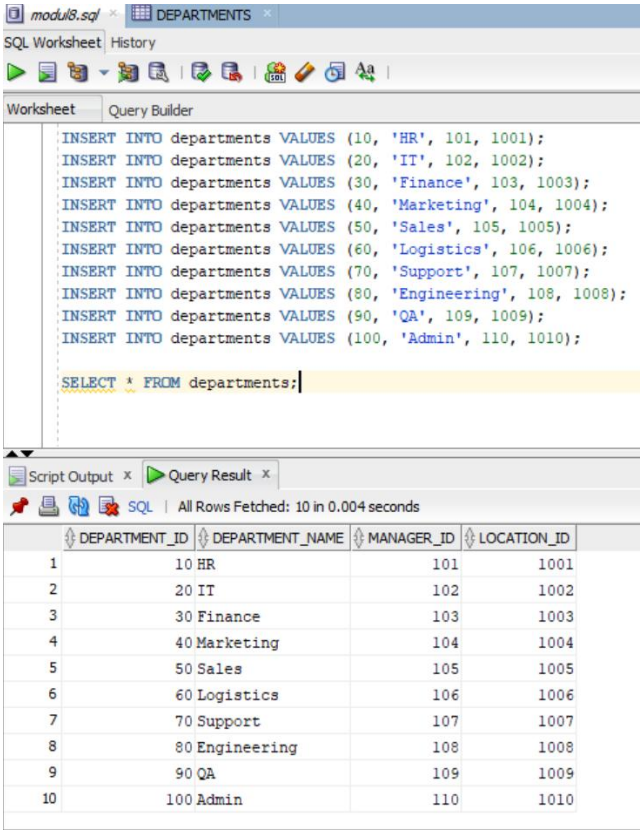
```
ANALYZE TABLE employees COMPUTE STATISTICS;
ANALYZE TABLE departments COMPUTE STATISTICS;
```

Task completed in 0.04 seconds

```
Table SUPPLIERS created.

Table PRODUCTS created.

Table EMPLOYEES analyzed.

Table DEPARTMENTS analyzed.
```

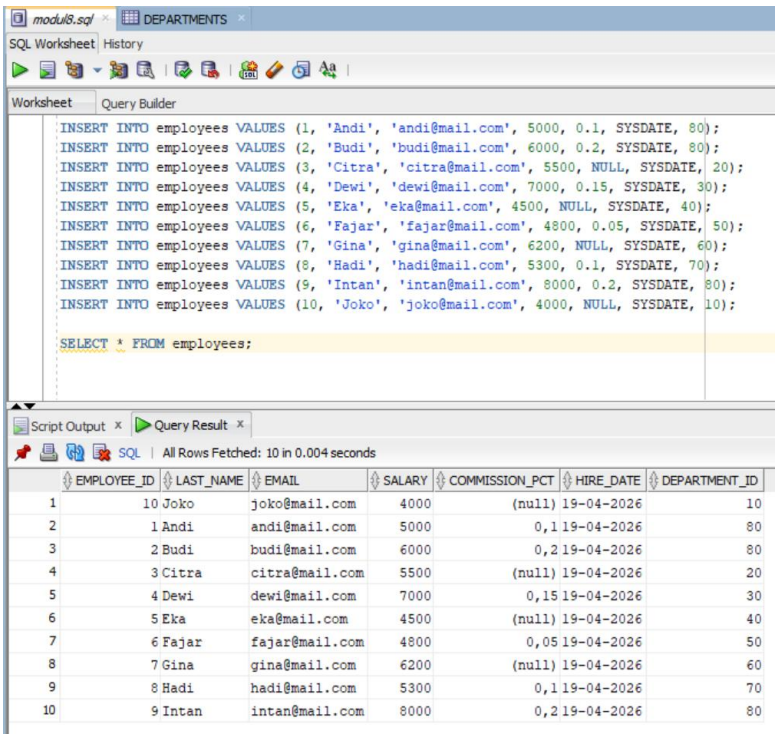
3. Isi data table

Tabel Departments



	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	HR	101	1001
2	20	IT	102	1002
3	30	Finance	103	1003
4	40	Marketing	104	1004
5	50	Sales	105	1005
6	60	Logistics	106	1006
7	70	Support	107	1007
8	80	Engineering	108	1008
9	90	QA	109	1009
10	100	Admin	110	1010

Tabel employees



	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPARTMENT_ID
1	10	Joko	joko@mail.com	4000	(null)	19-04-2026	10
2	1	Andi	andi@mail.com	5000	0,1	19-04-2026	80
3	2	Budi	budi@mail.com	6000	0,2	19-04-2026	80
4	3	Citra	citra@mail.com	5500	(null)	19-04-2026	20
5	4	Dewi	dewi@mail.com	7000	0,15	19-04-2026	30
6	5	Eka	eka@mail.com	4500	(null)	19-04-2026	40
7	6	Fajar	fajar@mail.com	4800	0,05	19-04-2026	50
8	7	Gina	gina@mail.com	6200	(null)	19-04-2026	60
9	8	Hadi	hadi@mail.com	5300	0,1	19-04-2026	70
10	9	Intan	intan@mail.com	8000	0,2	19-04-2026	80

Tabel Suppliers

```sql
INSERT INTO suppliers VALUES (1, 'Supplier A', '0811');
INSERT INTO suppliers VALUES (2, 'Supplier B', '0812');
INSERT INTO suppliers VALUES (3, 'Supplier C', '0813');
INSERT INTO suppliers VALUES (4, 'Supplier D', '0814');
INSERT INTO suppliers VALUES (5, 'Supplier E', '0815');
INSERT INTO suppliers VALUES (6, 'Supplier F', '0816');
INSERT INTO suppliers VALUES (7, 'Supplier G', '0817');
INSERT INTO suppliers VALUES (8, 'Supplier H', '0818');
INSERT INTO suppliers VALUES (9, 'Supplier I', '0819');
INSERT INTO suppliers VALUES (10, 'Supplier J', '0820');

SELECT * FROM suppliers;
```

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	SUPPLIER_ID	SUPPLIER_NAME	CONTACT
1	1	Supplier A	0811
2	2	Supplier B	0812
3	3	Supplier C	0813
4	4	Supplier D	0814
5	5	Supplier E	0815
6	6	Supplier F	0816
7	7	Supplier G	0817
8	8	Supplier H	0818
9	9	Supplier I	0819
10	10	Supplier J	0820

Tabel Products

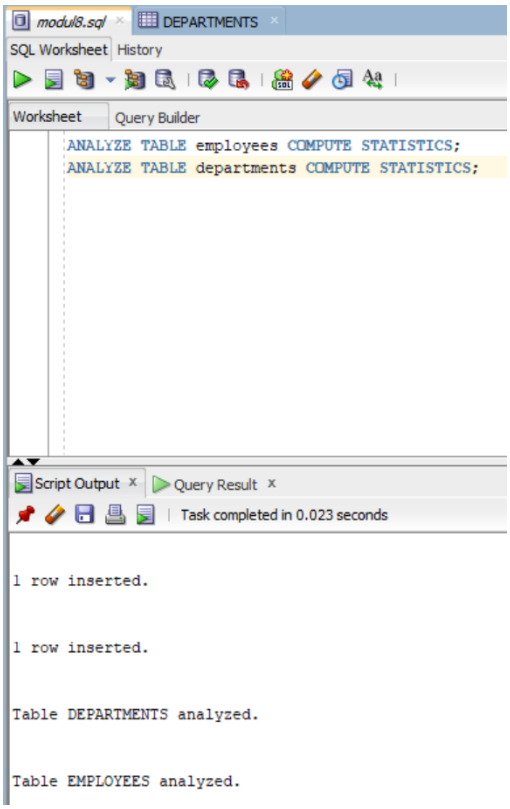
```sql
INSERT INTO products VALUES (1, 'Laptop', 10, 1);
INSERT INTO products VALUES (2, 'Mouse', 50, 2);
INSERT INTO products VALUES (3, 'Keyboard', 30, 3);
INSERT INTO products VALUES (4, 'Monitor', 20, 4);
INSERT INTO products VALUES (5, 'Printer', 15, 5);
INSERT INTO products VALUES (6, 'Scanner', 12, 6);
INSERT INTO products VALUES (7, 'Speaker', 25, 7);
INSERT INTO products VALUES (8, 'Router', 18, 8);
INSERT INTO products VALUES (9, 'Flashdisk', 60, 9);
INSERT INTO products VALUES (10, 'Harddisk', 22, 10);

SELECT * FROM products;
```

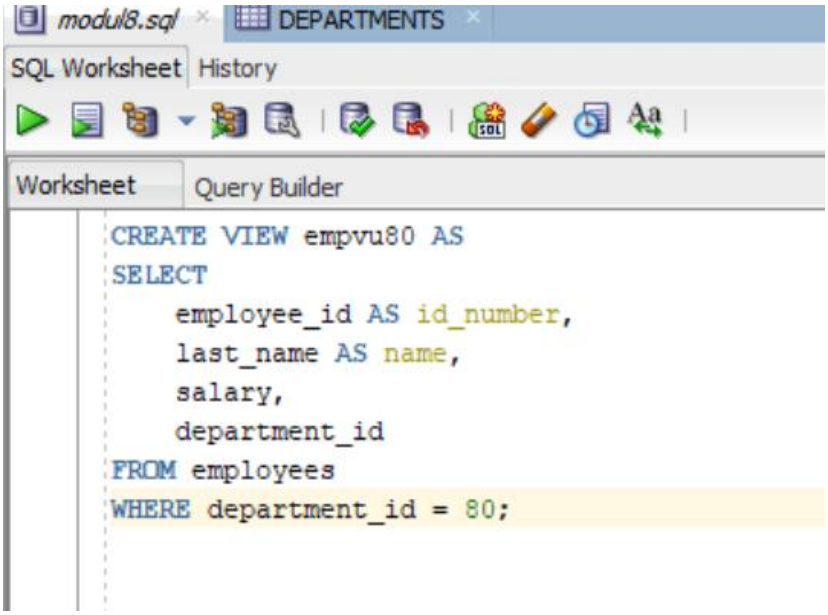
All Rows Fetched: 10 in 0.005 seconds

	PRODUCT_ID	PRODUCT_NAME	STOCK	SUPPLIER_ID
1	1	Laptop	10	1
2	2	Mouse	50	2
3	3	Keyboard	30	3
4	4	Monitor	20	4
5	5	Printer	15	5
6	6	Scanner	12	6
7	7	Speaker	25	7
8	8	Router	18	8
9	9	Flashdisk	60	9
10	10	Harddisk	22	10

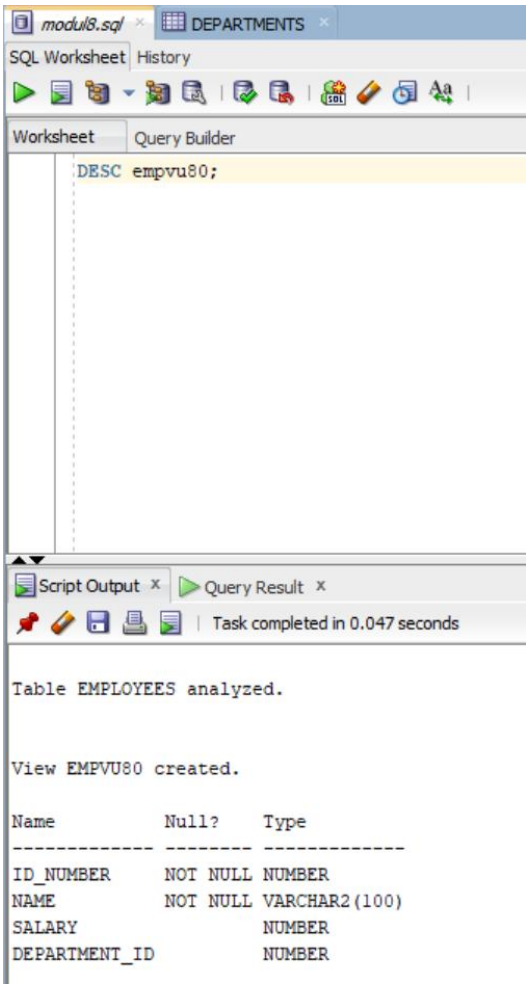
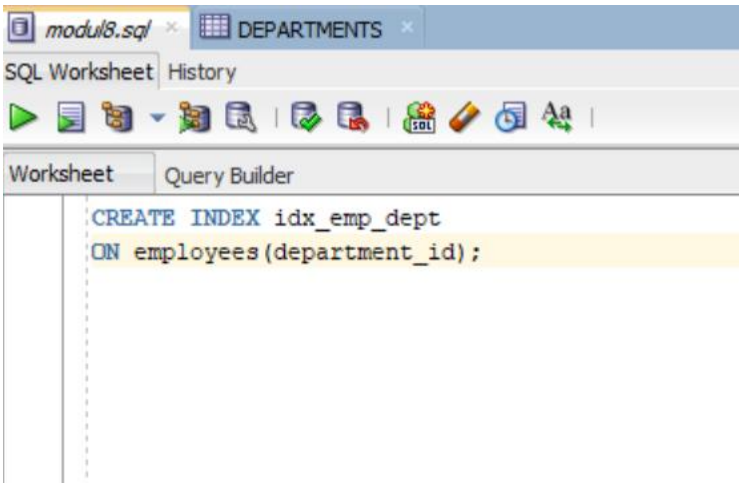
4. **Analisis statistic setelah isi data**



5. **Membuat VIEW empvu80**



6. Menampilkan Struktur VIEW



7. Membuat non unique index pada foreign key